

Pascal Jansen

Magirus-Deutz Strasse 4, 89077 Ulm, Germany — Research Associate, Ulm University, Ulm, Germany
pascal.j2@gmx.de — [Google Scholar](#) — [LinkedIn](#) — [GitHub](#) — pascal-jansen.github.io — [ORCID](#)

RESEARCH INTERESTS

Autonomous systems can be simulated, optimized, and stress-tested before deployment, yet the individual human evaluations that shape safety, trust, and acceptance often become observable only through situated interaction. Responses to a vehicle trajectory, robot motion, or explanation overlay are latent and context-dependent because people evaluate autonomous behavior through concrete exposure under uncertainty. My work addresses this methodological limitation by augmenting autonomous-system design with simulations and models of individual human experience. I develop human-in-the-loop optimization methods, adaptive interfaces, and mixed-reality study platforms that use subjective, behavioral, and physiological feedback as design guidance before real-world deployment. Across automated mobility, service and humanoid robotics, and extended reality, this research makes individual human evaluations computationally usable for designing autonomous systems that are safer, more understandable, and accepted in situated use.

EDUCATION

Ulm University, Ulm, Germany 06/2021 — present
Doctor of Science (PhD) Candidate in Human-Computer Interaction
Thesis Title: *A Framework for Computational User-Centered Optimization of Human-Vehicle Interaction Design*
Advisor: [Prof. Dr. Enrico Rukzio](#)
Committee: [Prof. Dr. Antti Oulasvirta](#) (Aalto University, Finland), [Prof. Dr. Philipp Wintersberger](#) (IT:U Linz, Austria)
Planned Defense: 09/2026

University College London, London, UK 01/2025 — 03/2025
Visiting Research Scholar with [Prof. Dr. Mark Colley](#)

Ulm University, Ulm, Germany 04/2018 — 05/2021
Master of Science in Computer Science Overall Grade: 1.1, with Distinction - A-equivalent
Thesis Title: *SwiVR-Car-Seat: Utilizing a Motorized Swivel Seat to Explore Effects of Vehicle Motion on Interaction Quality in Virtual Reality Automated Driving* Grade: 1.0 - A-equivalent

Ulm University, Ulm, Germany 10/2014 — 03/2018
Bachelor of Science: Computer Science
Specialized Subject Area: Biology

JOB EXPERIENCE

Ulm University Ulm, Germany
Research Associate 06/2021 — present
I significantly contribute to research and teaching. Also, I played a key role in co-leading and implementing the externally funded project *SituWare*. Furthermore, I successfully endorsed novel topics in immersive data visualization and computational methods for inclusion in the department's research scope. My work incorporates numerous research approaches such as experiments, interviews, focus groups, literature reviews, observation, usability testing, & dataset creation. It has led to over **20 peer-reviewed full papers**.

Zefwih GbR Neu-Ulm, Germany
CEO & Co-Founder 01/2022 — present
At [Zefwih](#), I lead business and research strategy while contributing to developing desktop, mobile, and virtual reality user-centered applications for commercial, cultural, and research scopes. I also consult on early-stage digitalization projects, leveraging iterative design processes and HCI principles to encourage sustainable innovation. This experience refined my skills in HCI research from an industrial perspective, interaction design consultation, and project collaboration.

Ulm University Ulm, Germany
Student Research Assistant, Institute of Media Informatics 06/2019 — 02/2021
I contributed to the ideation and development of a multi-user, shared-content augmented-reality platform and eye-tracking applications for virtual reality. This involved literature research, designing studies, and contributing to multiple publications. I also supported multiple lectures and exercises, tutored students during course projects, and evaluated their coursework.

Ulm University Ulm, Germany
Student Research Assistant, Institute of Communications Engineering 12/2018 — 03/2019
I led the ideation and development of voice-assistance applications in dialogue systems engineering, conducted literature research, and supported publication work. Additionally, I actively participated in internal review processes.

GRANTS, ACADEMIC AND INDUSTRIAL SUPPORT

Personal and Project Funding

- *Startup Your Career* support (10.000€) by the Graduate & Professional Training Center, Ulm University
- Supported by the Mobility Program of Ulm University (2.800€)
- Supported by the German Federal Ministry of Transport and Digital Infrastructure (25.000€)

Contributions to Funded Projects

- Contributed conceptual input that informed the successful funding proposal of the Deutsche Forschungsgemeinschaft - Reinhart Koselleck Project *VRooms: Fighting Climate Change with Increasing the Utilization of Buildings by Virtualizing Rooms through Everyday Extended Reality* ([DFG project link](#))

SERVICE and VOLUNTEERING ACTIVITIES

Associate Chair / Program Committee Member

- Full Paper - AutomotiveUI '24'25'26, MuC '24'25, CHIWORK '25'26, **CHI '26**
- Member of the **Best Paper Award Committee** for **CHI '26**
- Late-Breaking Works (LBW) - CHI '23'24'25

Organizing Committee

- [AutomotiveUI 2023](#) (Registration Chair)
- [MuC 2026](#) (Student Research Competition Chair)
- Co-Organizer of the [Post-CHI Summer School On Automotive User Interfaces and Future Mobility](#)
- Co-Organizer of the [German Pre-CHI 2022](#) hosted in Ulm, Germany with more than 70 participants

Editorial and Journal Service

- **ACM Transactions on Computer–Human Interaction (TOCHI)** – Distinguished Reviewer Board (since 2026)

Peer Reviewing

Over 250 peer-reviews completed so far for AutomotiveUI, VRST, **CHI**, IDC, ICWSM, GLS, MobileHCI, IMX, ETRA, COGAIN, CogSci, DIS, EICS PACM, CHI Play, ISS, ISMAR, Creativity & Cognition, MuC, NordiCHI, ICIS, OzCHI, ISWC, IEEE VIS, ICMI, HAI, PacificVis, **CSCW**, **UIST**, TEI, **IMWUT**, **TVCG**, **TRF**, and BIT

Student Volunteer: CHI 2023, MuC 2025

TEACHING

Design Thinking for Interactive Systems

Course Organizer: Co-organized an interdisciplinary project emphasizing user-centered design and design thinking, undertaken as a year-long, research-driven group effort culminating in several publications. Fall 2021 — Spring 2025

User Interface Software Technologies

Course Organizer and Lecturer: Led the development of course materials and delivered a weekly (interactive) lecture on designing user interfaces and interaction technologies, emphasizing a user-centered design process. Starting from the basics of interactive systems, the course explored interface creation, different media, and formal methods—particularly HCI notations. I co-led the evaluation process by grading the final exam and ensuring a fair, thorough assessment of student performance. Spring 2022 — Spring 2024

Automotive User Interfaces and Interactive Vehicle Applications

Course Organizer and Lecturer: Led the development of course materials and held a weekly (interactive) lecture on practical insights on developing and researching automotive user interfaces, teaching the user-centered design process. Delivered a yearly in-depth lecture on future mobility research and computational methods for automated vehicle user interface design. Co-led the evaluation process by grading the final exam and ensuring a fair and thorough assessment of student performance. Fall 2021 — Fall 2025

Research Trends in Media Informatics

Course Organizer: Co-organized the course, mentored PhD students on course structure and content, and provided detailed one-on-one instruction to over 10 students in conducting literature surveys using the [PRISMA](#) method. Additionally, took an active role in assessing and grading students' work. Fall 2021 — Fall 2024

Technical Foundations of Media Development

Lecturer and Course Contributor: Supported continuing-education students in developing technical foundations for interactive systems design, including software development, prototyping workflows, debugging, and interaction implementation. The course required accessible explanations for students with heterogeneous professional backgrounds while maintaining the technical and methodological standards expected in media informatics and HCI. Fall 2023 — Spring 2026

THESIS SUPERVISION (Excerpt, Total>30)

Bachelor Theses

- Tim Eckstein (Ulm University; 2026)
- Hermann Fröhlich (Ulm University and [PlanB. GmbH](#); 2024)
- Nam Anh Le Cong (Ulm University; 2024)
- Julius Schürle (Ulm University; 2023)
- Benno Hölz (Ulm University; 2023)
- Oliver Schmid (Ulm University; 2022)

Master Theses

- [Julian Britten](#) (Ulm University; 2025) **now PhD student at Ulm University**
- Sepide Ansari (Ulm University and [Spiegel Institute Mannheim GmbH](#); 2025)
- Mugdha Keskar (Ulm University; 2024)
- Svenja Krauß (Ulm University; 2023)
- Alexander Häusele (Ulm University; 2023)
- Alexandra Nick (Ulm University and [Cerence GmbH](#); 2023)

INVITED TALKS and LECTURES

- [TU Hamburg Institute for Ethics in Technology](#) (08.06.2026): "Low-Cost High Fidelity Simulators for Interaction with Future Mobility"; online; invited by [Jan Hölzer](#)
- [UCL Interaction Centre](#) (04.03.2026): "From One-Size-Fits-All to Preference-Based Interactions with Technology"; *Future Interfaces Course*; in-person; invited by [George Chalhoub](#)
- [UCL Interaction Centre](#) (13.02.2025): "Ubiquitous Personalization"; *Future Interfaces Course*; in-person; invited by [Mark Colley](#)

AWARDS and NOMINATIONS

Outstanding Reviewer Recognition: ISS '22, AutomotiveUI '22 '23 '25, DIS '23, IEEE VIS '23, CHI '24 LBW, CSCW '24, MuC '24, UIST '24 '25 '26, IMWUT '22 '24 '25 '26, and CHI '24 (x2) '25 '26

I have received **20 Outstanding Reviewer Recognitions**, demonstrating my commitment to excellence in academic review processes. These recognitions reflect my deep understanding and critical thinking skills, which enable me to advance scholarly discourse.

Honorable Mention Award at MUM '25 - DOI: [10.1145/3771882.3771888](#)

This work advances flexible, space-efficient interior design through *AirClick*, a modular system of interactive inflatables that enables rapid, on-demand room transformation. In a lab study (N=20) across four scenarios, participants successfully reconfigured spaces using the *AirClick* modules. Findings demonstrate suitability for everyday environments with low to medium robustness demands, highlighting *AirClick*'s potential for customizable, space-saving room adaptation.

Honorable Mention Award at CHI '25 - DOI: [10.1145/3706598.3713514](#)

This paper addressed the challenge of scalable automotive user interface design. We implemented *OptiCarVis*, a system integrating Human-in-the-Loop Bayesian Optimization. An online study (N=117) demonstrates the efficacy of *OptiCarVis* in significantly improving trust, acceptance, perceived safety, and predictability, without increasing cognitive load.

Audience Choice Award at CHI PLAY '20 - DOI: [10.1145/3383668.3419917](#)

As technical attacks become increasingly difficult to execute, social engineering exploits the human element in information security, often going undetected and resulting in high organizational costs. To address this, I led the ideation and development of "The Social Engineer", an immersive VR game designed to raise awareness by letting players experience various social engineering methods in a simulated company environment. This extensible tool can serve as employee self-training, aid security experts in teaching awareness, and provide consumer engagement.

Young Talent Award Nominee at German Computer Game Awards '21

With "The Social Engineer", I was nominated for the Young Talent Award at the German Computer Game Awards '21 — the largest national competition for digital games in Germany. The same project was also a finalist for the **Best Student Game Award** at the internationally recognized [Games for Change Awards '22](#), and was subsequently nominated for the [nextReality.Contest '22](#) in the "Young Talent" category, a major German competition for immersive media innovation.

SKILLS

- **Research:** Proficient in **quantitative analysis** using R (parametric/non-parametric data, linear regression, hierarchical models) and **qualitative analysis** following [Saldaña's](#) guidelines
- **Coding:** R (data analysis, visualization, web scraping), Python (computational modeling), Java, and C# (particularly with **Unity**)
- **Fabrication & Engineering:** Experienced in 3D printing, electrical engineering, and microcontroller programming (Arduino, Raspberry Pi)
- **Sensing Technologies:** Adept at using physiological sensors (eye-tracking on desktop/head-mounted display, wearable sensors for heart rate and skin conductance), motion tracking (computer vision using RGB camera input and infrared-based body/hand tracking), and vehicle telemetry (interior/exterior cameras, OBD) for data analysis and interactive, as well as adaptive systems
- **Open Science:** Proficient in [open science practices](#), transparency, reproducibility, and accessibility; experienced in sharing data ([example](#)) via open-access platforms, using open-source tools, and contributing to community projects
- **Languages:** German (native), English (proficient), French (intermediate).

MAJOR PUBLICATIONS (CHI, IMWUT, UIST, TOCHI)

ACM [CHI/IMWUT/UIST/TOCHI](#) are widely recognized as the premier venues for publishing research in the field of HCI. They are highly competitive, with acceptance rates typically ranging between 20-25%.

CHI

1. **P. Jansen***, J. Britten*, M. Colley*, M. Sasalovici, & E. Rukzio, MIRAGE: Enabling Real-Time Automotive Mediated Reality
In Proc. of CHI 2026, ACM, conditionally accepted *Joint First Authors, doi: [2601.19385](#)
2. J. Susak, Y. Liu, **P. Jansen***, & M. Colley, ProVoice: Designing Proactive Functionality for In-Vehicle Conversational Assistants using Multi-Objective Bayesian Optimization to Enhance Driver Experience
In Proc. of CHI 2026, ACM, conditionally accepted, doi: [2601.19421](#)
3. M. Colley, S. Kopp, D. Dey, **P. Jansen***, & E. Rukzio, eHMI for All - Investigating the Effect of External Communication of Automated Vehicles on Pedestrians, Manual Drivers, and Cyclists
In Proc. of CHI 2026, ACM, conditionally accepted, doi: [2601.19440](#)
4. **P. Jansen***, M. Colley*, S. Krauß, D. Hirschle, & E. Rukzio, OptCarVis: Improving Automated Vehicle Functionality Visualizations Using Bayesian Optimization to Enhance User Experience
In Proc. of CHI 2025, ACM, doi: [10.1145/3706598.3713514](#), *Joint First Authors
CHI Honorable Mention Award for Best Paper (top 5%)
5. M. Colley*, **P. Jansen***, M. Keskar, & E. Rukzio, Improving External Communication of Automated Vehicles Using Bayesian Optimization
In Proc. of CHI 2025, ACM, doi: [10.1145/3706598.3714187](#), *Joint First Authors

6. M. Sasalovici, A. Zeqiri, R. C. Schramm, O. J. A. Nuñez, **P. Jansen**, J.P. Freiwald, M. Colley, C. Winkler, & E. Rukzio, Bumpy Ride? Understanding the Effects of External Forces on Spatial Interactions in Moving Vehicles
In Proc. of CHI 2025, ACM, doi: [10.1145/3706598.371407](https://doi.org/10.1145/3706598.371407)
7. L. Meinhardt, C. Schramm, **P. Jansen**, M. Colley, & E. Rukzio, Fly Away: Evaluating the Impact of Motion Fidelity on Optimized User Interface Design via Bayesian Optimization in Automated Urban Air Mobility Simulations
In Proc. of CHI 2025, ACM, doi: [10.1145/3706598.3713288](https://doi.org/10.1145/3706598.3713288)
8. A. Zeqiri, J. Britten, C. Schramm, **P. Jansen**, M. Rietzler, & E. Rukzio, PlantPal: Leveraging Precision Agriculture Robots to Facilitate Remote Engagement in Urban Gardening
In Proc. of CHI 2025, ACM, doi: [10.1145/3706598.3713180](https://doi.org/10.1145/3706598.3713180)
9. M. Colley, B. Wanner, M. Rädler, M. Rötzer, J. Frommel, T. Hirzle, **P. Jansen**, & E. Rukzio, Effects of a Gaze-Based 2D Platform Game on User Enjoyment, Perceived Competence, & Digital Eye Strain
In Proc. of CHI 2024, ACM, doi: [10.1145/3613904.3641909](https://doi.org/10.1145/3613904.3641909)
10. **P. Jansen**, J. Britten, A. Häusele, T. Segschneider, M. Colley, & E. Rukzio, AutoVis: Enabling Mixed-Immersive Analysis of Automotive User Interface Interaction Studies
In Proc. of CHI 2023, ACM, doi: [10.1145/3544548.3580760](https://doi.org/10.1145/3544548.3580760), [Website Link]

IMWUT

1. A. Zeqiri, **P. Jansen**, J. Rixen, M. Rietzler, and E. Rukzio, 'Eco Is Just Marketing': Unraveling Everyday Barriers to the Adoption of Energy-Saving Features in Major Home Appliances
In Proc. IMWUT 2024, ACM, doi: [10.1145/3643558](https://doi.org/10.1145/3643558)
2. **P. Jansen**, M. Colley, & E. Rukzio, A Design Space for Human Sensor and Actuator Focused In-Vehicle Interaction Based on a Systematic Literature Review
In Proc. IMWUT 2022, ACM, doi: [10.1145/3534617](https://doi.org/10.1145/3534617)
3. M. Colley, **P. Jansen**, E. Rukzio, and J. Gugenheimer, SwiVR-Car-Seat: Exploring Vehicle Motion Effects on Interaction Quality in Virtual Reality Automated Driving Using a Motorized Swivel Seat
In Proc. IMWUT 2021, ACM, doi: [10.1145/3494968](https://doi.org/10.1145/3494968)

UIST

- **P. Jansen**, F. Fischbach, J. Gugenheimer, E. Stemasov, J. Frommel, and E. Rukzio, ShARe: Enabling Co-Located Asymmetric Multi-User Interaction for Augmented Reality Head-Mounted Displays
In Proc. UIST 2020, ACM, doi: [10.1145/3379337.3415843](https://doi.org/10.1145/3379337.3415843)

TOCHI

- T. Hirzle, F. Fischbach, J. Karlbauer, **P. Jansen**, J. Gugenheimer, E. Rukzio, and A. Bulling, Understanding, Addressing, and Analysing Digital Eye Strain in Virtual Reality Head-Mounted Displays
In ACM Transactions on Computer-Human Interaction (TOCHI) 2022, ACM, doi: [10.1145/3492802](https://doi.org/10.1145/3492802)

FURTHER PUBLICATIONS

Journal Paper

[Transportation Research Part F](#), with an impact factor of 4.60 (2022) is considered to be top-tier journals in traffic psychology.

1. **P. Jansen***, M. Colley*, T. Pfeifer, & E. Rukzio, Visualizing Imperfect Situation Detection and Prediction in Automated Vehicles: Understanding Users
In Transportation Research Part F: Traffic Psychology and Behaviour 2024, Elsevier, *Joint First Authors, doi: [10.1016/j.trf.2024.05.015](https://doi.org/10.1016/j.trf.2024.05.015)
2. **P. Jansen***, M. Colley*, Max Rädler*, Jonas Schwedler, & E. Rukzio, Longitudinal effects of visualizing uncertainty of situation detection and prediction of automated vehicles on user perceptions
In Transportation Research Part F: Traffic Psychology and Behaviour 2024, Elsevier, *Joint First Authors, doi: [10.1016/j.trf.2025.05.013](https://doi.org/10.1016/j.trf.2025.05.013)

Conference Full Paper

1. T. Drey, F. Fischbach, **P. Jansen**, J. Frommel, M. Rietzler, and E. Rukzio, To Be or Not to Be Stuck, or Is It a Continuum?: A Systematic Literature Review on the Concept of Being Stuck in Games
In Proc. CHI PLAY 2021, ACM, doi: [10.1145/3474656](https://doi.org/10.1145/3474656)
2. M. Kraus, F. Fischbach, **P. Jansen**, and W. Minker, A comparison of explicit and implicit proactive dialogue strategies for conversational recommendation
In Proc. of the Twelfth Language Resources and Evaluation Conference (LREC) 2020, ACM, urn: [urn:nbn:de:bvb:384-opus4-1013633](https://nbn-resolving.org/urn:nbn:de:bvb:384-opus4-1013633)
3. M. Raab, L. Miller, Z. Zeng, **P. Jansen**, M. Baumann, and J. Kraus, Assessing Pedestrian Behavior Around Autonomous Cleaning Robots in Public Spaces: Findings from a Field Observation
34th IEEE Int. Conf. on Robot and Human Interactive Communication (RO-MAN) 2025, IEEE, doi: [RO-MAN63969.2025.11217688](https://doi.org/10.1109/RO-MAN63969.2025.11217688)
4. **P. Jansen**, B. Hölz, J. Britten, M. Colley, and E. Rukzio, AirClick: Modularized Interactive Inflatables for On-Demand Room Transformation
In Proc. of the International Conference on Mobile and Ubiquitous Multimedia (MUM) 2025, ACM, doi: [10.1145/3771882.3771888](https://doi.org/10.1145/3771882.3771888)
Honorable Mention Award for Best Paper (top 5%)

Conference Short Paper

1. M. Colley, J. Czymmek, **P. Jansen**, L.-M. Meinhardt, P. Ebel, & E. Rukzio, UAM-SUMO: Simulacra of Urban Air Mobility Using SUMO To Study Large-Scale Effects
In Proc. HRI 2025, ACM, doi: [10.5555/3721488.3721610](https://doi.org/10.5555/3721488.3721610), [GitHub Link](#)
2. **P. Jansen***, M. Colley*, E. Wimmer, J. Maresch, & E. Rukzio, HUD-SUMO: Simulacra of In-Vehicle Head-Up Displays Using SUMO To Study Large-Scale Effects
In Proc. HRI 2025, ACM, doi: [10.5555/3721488.3721614](https://doi.org/10.5555/3721488.3721614), [GitHub Link](#); * Joint First Authors
3. M. Colley, J. Czymmek, M. Küçükocak, **P. Jansen**, & E. Rukzio, PedSUMO: Simulacra of Automated Vehicle-Pedestrian Interaction Using SUMO To Study Large-Scale Effects
In Proc. HRI 2024, ACM, doi: [10.1145/3610977.3637478](https://doi.org/10.1145/3610977.3637478), [GitHub Link](#)
4. **P. Jansen** and F. Fischbach, The Social Engineer: An Immersive Virtual Reality Educational Game to Raise Social Engineering Awareness
In Proc. CHI PLAY EA 2020, ACM, doi: [10.1145/3383668.3419917](https://doi.org/10.1145/3383668.3419917)
Audience Choice Award

Workshop

1. M. Haimlerl, **P. Jansen**, A. Riener, and M. Colley, Accessible Automated Automotive Workshop Series (A3WS): Focus Accessibility in Mobility
In Proc. MuC EA 2025, Gesellschaft für Informatik e.V, doi: [10.18420/muc2025-mci-ws-104](https://doi.org/10.18420/muc2025-mci-ws-104)
2. **P. Jansen** and M. Colley, Human-in-the-Loop Bayesian Optimization for ALL: Practical Applications in Human-Computer Interaction Challenges
In Proc. MuC EA 2025, Gesellschaft für Informatik e.V, doi: [10.18420/muc2025-mci-tut-152](https://doi.org/10.18420/muc2025-mci-tut-152)

Extended Abstracts

1. T. Drey, **P. Jansen**, F. Fischbach, J. Frommel, & Enrico Rukzio, Towards Progress Assessment for Adaptive Hints in Educational Virtual Reality Games
In Proc. CHI EA 2020, ACM, doi: [10.1145/3334480.3382789](https://doi.org/10.1145/3334480.3382789)
2. L.-M. Meinhardt, M. Colley, A. Balci, A. Degenhard, **P. Jansen**, & E. Rukzio, Rethinking Simulator Fidelity in Urban Air Mobility: A Null-Result Investigation into the Effects of Induced Arousal and Motion Cues
In Proc. CHI EA 2026, ACM, doi: [10.1145/3772363.3798420](https://doi.org/10.1145/3772363.3798420)

Demo

- **P. Jansen**, J. Britten, A. Häusele, T. Segschneider, M. Colley, & E. Rukzio, A Demonstration of AutoVis: Enabling Mixed-Immersive Analysis of Automotive User Interface Interaction Studies
In Proc. AutoUI EA 2023, ACM, doi: [10.1145/3581961.3610374](https://doi.org/10.1145/3581961.3610374)

Workshop Position Paper

1. **P. Jansen**, Human-in-the-Loop Optimization for Inclusive Design: Balancing Automation and Designer Expertise
CHI 2025 Workshop Access InContext: Futuring Accessible Prototyping Tools and Methods. April 26, 2025. Yokohama, Japan, doi: [10.48550/arXiv.2505.08375](https://doi.org/10.48550/arXiv.2505.08375)
2. **P. Jansen**, From Human Negotiation to Agent Negotiation: Personal Mobility Agents in Automated Traffic
CHI 2026 Workshop AutomationXP26: Agentic Automation Experiences – Rethinking the Interaction of Humans and AI Agents. April 14, 2026. Barcelona, Spain, doi: [10.48550/arXiv.2603.01035](https://doi.org/10.48550/arXiv.2603.01035)
3. **P. Jansen**, Toward Governing Perception in Safety-Critical Mediated Reality on the Move
CHI 2026 Workshop Next Steps for Augmented Reality On-the-Move: Challenges & Opportunities. April 14, 2026. Barcelona, Spain, doi: [10.48550/arXiv.2603.08138](https://doi.org/10.48550/arXiv.2603.08138)